ModelIt!

Teacher's Guide

Why Does a Glow Stick Stop Glowing?

Modeling Chemical Kinetics and Energy Transformation in Chemiluminescence

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-PS1-4, HS-PS1-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS1-4, HS-PS1-6

Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy. Refine the design of a chemical system by specifying a change in conditions that would produce increased amounts of a product at equilibrium.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model of chemiluminescent reaction kinetics to predict how temperature and concentration changes affect light output intensity and duration.
Disciplinary Core Idea	**PS1.B Chemical Reactions / PS1.A Structure and Properties of Matter** — Chemical reactions involve the breaking and forming of bonds with associated energy changes; the rate of reactions depends on temperature, concentration, and molecular properties.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students trace energy transformation from chemical bond energy through molecular excitation to visible light emission, accounting for energy conservation and dissipation as heat.

CAST Testing Alignment

MULTIPLE CHOICE:

A student places identical glow sticks in hot water (60 degrees C) and cold water (5 degrees C). After 30 minutes, which glow stick has produced more total light?

CONSTRUCTED RESPONSE:

Using your model, explain why a glow stick placed in boiling water produces an intense burst of light that fades quickly, while one in the freezer produces a dim glow that lasts for days. Reference Chemical Concentration, Reaction Rate, and Product Buildup in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

A glow stick is a chemistry lesson in a plastic tube. When you snap it, you initiate a precisely engineered chemiluminescent reaction that converts chemical bond energy directly into visible light — no battery, no filament, no electricity. Understanding why it glows, why it stops, and why temperature matters reveals fundamental principles of chemical kinetics, energy transformation, and molecular dynamics.

The Chemistry Inside

A glow stick contains two separated chemicals: a solution of diphenyl oxalate (the energy source) and a fluorescent dye in the outer tube, and a glass vial of hydrogen peroxide (the oxidizer) in the center. When you snap the stick, the glass vial breaks and the chemicals mix. Hydrogen peroxide oxidizes diphenyl oxalate to produce phenol, CO_2, and a high-energy intermediate called 1,2-dioxetanedione. This intermediate decomposes and transfers its energy to the fluorescent dye, which emits a photon of visible light as it returns to its ground state. The color depends on the dye: green (9,10-bis(phenylethynyl)anthracene), blue (9,10-diphenylanthracene), red (rhodamine B).

Temperature and Reaction Rate

The Arrhenius equation describes how temperature affects reaction rate: $k = Ae^{(-Ea/RT)}$, where k is the rate constant, Ea is activation energy, R is the gas constant, and T is temperature in Kelvin. For most reactions, a 10 degree Celsius increase approximately doubles the reaction rate. This is because higher temperature means molecules move faster, collide more frequently, and a larger fraction exceeds the activation energy threshold. For a glow stick, this means doubling the temperature roughly doubles the light intensity — but also halves the duration.

The Brightness-Duration Trade-Off

A glow stick contains a fixed amount of chemical reactant — its 'fuel.' The total light energy it can produce is determined by this fuel amount, not by how fast it's consumed. Heating accelerates the reaction, consuming fuel faster and producing brighter light for a shorter time. Cooling slows the reaction, consuming fuel more slowly and producing dimmer light for a longer time. The total integrated light output is approximately the same. This is a direct consequence of conservation of energy — the chemical energy in the bonds is converted to light energy regardless of the rate.

Product Buildup and Reaction Slowdown

As the reaction proceeds, reactant concentration decreases and product concentration increases. According to the law of mass action, reaction rate is proportional to reactant concentration. As diphenyl oxalate and hydrogen peroxide are consumed, fewer reactant molecules are available to collide and react. Simultaneously, the accumulating products (phenol and CO_2) dilute the remaining reactants. This creates a characteristic exponential decay in light output — bright at first, then progressively dimmer. The glow stick doesn't stop abruptly; it fades because the reaction asymptotically approaches completion.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the chemiluminescent system: Chemical Concentration, Temperature, Reaction Rate, Fluorescent Dye Activation, Light Output, Energy Dissipation, and Product Buildup — with Temperature as the external variable.

E **Establish Relationships**

Students determine that Temperature drives Reaction Rate (Arrhenius relationship), which determines both Fluorescent Dye Activation (producing Light Output) and the rate of Chemical Concentration decline. Product Buildup slows the reaction as it progresses. Energy Dissipation represents losses.

V **Visualize & Model**

Students build a computational model showing how temperature controls the rate of chemiluminescent light production and the trade-off between intensity and duration.

E **Evaluate Outcomes**

Students run room temperature, hot water bath, and freezer scenarios to compare brightness curves, duration, and total light output, confirming the brightness-duration trade-off.

R **Revise & Extend**

Students add Catalyst Presence, Dye Quantum Yield, or Solvent Viscosity to explore how chemical formulation choices affect glow stick performance optimization.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Snap. Shake. Glow. No batteries, no electricity, no heat — just chemistry turning bond energy into light. But why does it stop? And why does heating it make it brighter but shorter?"*

Watch For

If possible, snap a glow stick in class. Place pieces in hot water and ice water. Observe in real time. If not possible, show a time-lapse video of the experiment.

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling one of the coolest chemical reactions you can hold in your hand — and discovering the fundamental rules that control ALL chemical reactions."*

Watch For

Have students read objectives. Pre-teach 'chemiluminescence' and 'activation energy.' Quick-write: Why do you think hot glow sticks are brighter?

Slide 3: Big Question

3 min

Say: *"A glow stick starts bright and fades to nothing. Is it running out of energy? Running out of chemicals? Both? And why does temperature change the whole story?"*

Watch For

Students hypothesize: What's happening inside the glow stick over time? Draw their mental model. Share and compare.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model a chemical reaction from start to finish — tracking every molecule, every photon, every degree of temperature."*

Watch For

Preview each LEVER step. Emphasize that this model connects molecular-level chemistry to observable macroscopic behavior.

Slide 5: Activity 1: Components

8 min

Say: *"One thing you control: temperature. The chemistry handles the rest. But how you set that one variable changes everything."*

Watch For

Guide component sorting. Discuss why Temperature is external (you control it with hot/cold water) while all chemical processes are internal responses.

Slide 6: Activity 2: Connections

8 min

Say: *"When temperature goes up, reaction rate goes up. But so does chemical consumption. Trace the consequences all the way to the last photon."*

Watch For

Help students map the cascade: Temperature drives Reaction Rate, which drives both Light Output and Chemical Concentration depletion. Product Buildup slows everything down.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three glow sticks, three temperatures. Which one is brightest? Which lasts longest? And here's the real question: which produces the most total light?"*

Watch For

Students predict brightness curves for all three temperatures before running simulations. Compare total light output (area under the curve) across scenarios.

Slide 8: Discoveries

5 min

Say: *"Conservation of energy just showed up in a glow stick. The total light is the same regardless of temperature — you're just choosing how to spend your chemical budget."*

Watch For

Lead discussion connecting glow stick behavior to conservation of energy. Discuss the Arrhenius equation in accessible terms.

Slide 9: STEM Challenge

12 min

Say: *"Search and rescue teams need three different glow sticks: a flare, a lantern, and a trail marker. Same chemistry, different specs. Design all three."*

Watch For

Groups optimize light stick formulations for three different emergency scenarios. Must specify concentration, temperature management, and expected performance. Present designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Temperature
- **INTERNAL:** Chemical Concentration, Reaction Rate, Fluorescent Dye Activation, Light Output, Energy Dissipation, Product Buildup

Relationships

- **Temperature to Reaction Rate:** POSITIVE (+) — Higher Temperature increases the kinetic energy of molecules, causing more frequent and more energetic collisions. According to the Arrhenius equation, a 10 degree Celsius increase roughly doubles the fraction of molecules exceeding activation energy, approximately doubling the Reaction Rate.
- **Reaction Rate to Chemical Concentration:** NEGATIVE (-) — A faster Reaction Rate consumes reactant molecules more rapidly, decreasing Chemical Concentration. This is a direct mass balance relationship — every reaction event converts one set of reactant molecules into products, reducing the pool of available reactants.
- **Product Buildup to Reaction Rate:** NEGATIVE (-) — As Product Buildup increases, the concentration of products rises relative to reactants. This dilutes the remaining reactants and shifts the system toward equilibrium, progressively slowing the net forward Reaction Rate according to the law of mass action.

Key Discoveries

- Temperature is the master controller of reaction rate — doubling the temperature approximately doubles the rate because more molecules exceed the activation energy threshold
- There is a fundamental trade-off between brightness and duration: heating a glow stick makes it brighter but shorter-lived because the same chemical fuel is consumed faster
- Product Buildup creates a negative feedback loop — as products accumulate, they dilute the reactants and slow the reaction, causing the gradual dimming observed in all glow sticks
- Total light output is roughly constant regardless of temperature — a hot glow stick and a cold glow stick convert approximately the same total chemical energy to light, just at different rates

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Glow sticks use electricity or radioactivity"	Glow sticks produce light entirely through chemical energy — no batteries, electricity, or radioactive materials are involved. The light comes from fluorescent dye molecules that are excited by energy from a chemical reaction (oxidation of diphenyl oxalate by hydrogen peroxide). This is chemiluminescence — the direct conversion of chemical bond energy to visible light photons.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heating a glow stick gives it more energy"	Heating a glow stick does not add energy to the chemical reaction — it increases the RATE at which existing chemical energy is released. The total chemical energy is determined by the amount of reactant, not the temperature. A hot glow stick converts its chemical fuel to light faster, producing brighter but shorter-lived light. A cold one converts the same fuel slower. Conservation of energy means the total light output is approximately the same.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"When the glow stick stops glowing, the chemicals are gone"	The chemicals are not gone — they have been transformed. The reactants (diphenyl oxalate and hydrogen peroxide) have been converted to products (phenol and CO2). The atoms are still there; they are just arranged in different molecules that store less energy. This is conservation of mass in action. The glow stops because the products don't have the high-energy bonds needed to excite the fluorescent dye.